

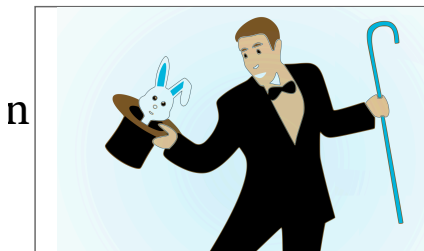
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Causal complexity: the missing piece in the Relevance puzzle

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Problem: Some events are exciting, not because they are simple, but because they seem incredible.

just) and measure the
s occurrence.



it magic, the
of the event (which is
nal situation) is huge.
the complexity gap,
ness.

As we did for cognitive complexity, we consider a resource-bounded notion of causal complexity. **Causal complexity**, also called **world complexity**, is defined as the shortest description (in bits) of a causal procedure that a human individual could reach at time t to account for the occurrence of a situation s . In a semi-formal way:

Causal complexity

$$C_w(s) = \min_p \{ \text{length}(p); W^t(p) = s \}$$

Here, W represents the observer's causal model of the world. W^t represents the same model that is allowed to run during t units of time. p is any procedure that respects causality as understood by the observer and that is available to W^t . Note that $C_w(s)$ may vary through time.

In many cases (e.g. when you meet a colleague on the way to the workplace), there is one single obvious scenario that only demands parameters to be set to precise values (such as time of departure from home). If so, causal complexity amounts to the description of these values.

Causal Complexity

1/1 point (ungraded)

In 1991, the New Jersey three-digit number lottery issued the same winning number, 741, twice in a row. What is the causal complexity of the event?

☒ 10 bits.

☐ 20 bits.



Answer

Correct:

Both answers are correct. The causal complexity of the event "741 twice" is about 20 bits, while the causal complexity of the event "same number twice" is about 10 bits (as the "world" needs 10 bits to adjust the second draw to the value of the first draw, whatever that value).

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✔ Correct (1/1 point)

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